Ethereum wallet behaviour is best treated as layered

- Goal: segment Ethereum EOA wallets into defensible behavioural roles.

- Main result: static roles, temporal rhythms, activity states, and network position overlap but do not collapse into one taxonomy.

- Academic contribution: a validation workflow, not just a clustering output.

The project moved from application idea to empirical foundation

- Initial idea: AI-governed crypto-bank simulator with agents and explainable decisions.

- Pivot: real Ethereum wallet behaviour needed to be understood before simulation assumptions could be trusted.

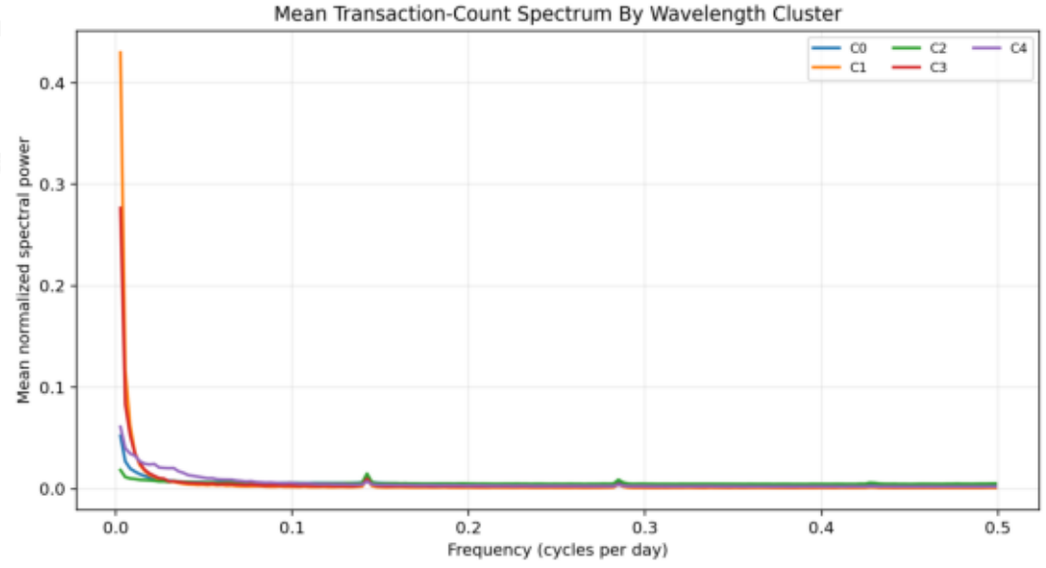
- Final direction: derive behavioural roles from data, then validate them with independent evidence.

Temporal signal analysis adds a second evidence layer

- Built daily 2022 wallet signals for a 10,000-wallet edge-aware sample.

- Frequency-domain features capture rhythms in transa

- Time-domain features capture active days, inactive ga
  and switching.

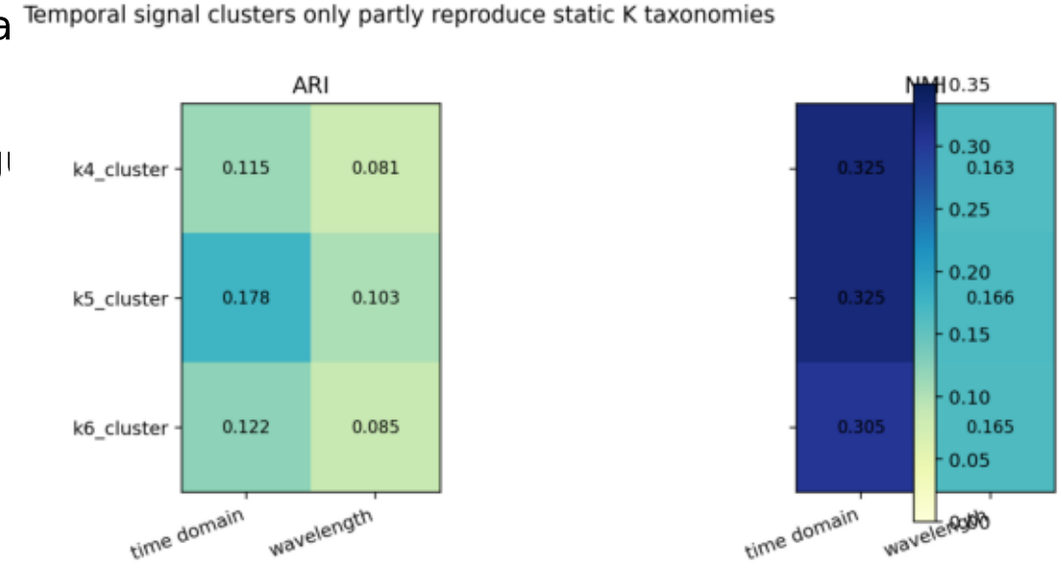


Signal clusters are structured but do not simply reproduce K5

- K5 has the highest correspondence with signal clusters among K4/K5/K6.

- The agreement remains low to moderate, so the signa
  duplicate of static roles.

- This gives the paper a stronger layered-behaviour arg

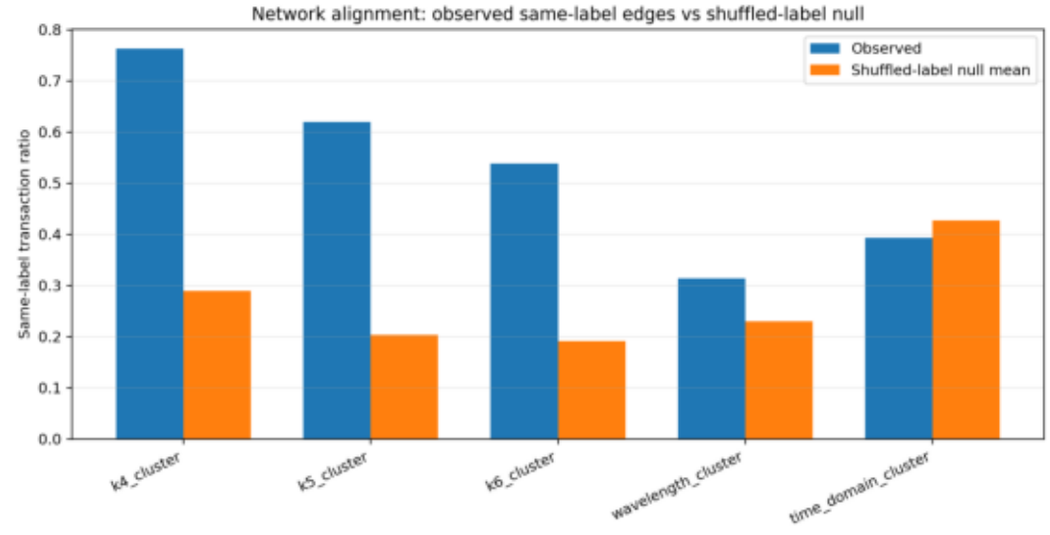


Network validation separates stronger and weaker claims

- Static K taxonomies show strong same-label transaction concentration.

- Wavelength clusters also exceed the shuffled-label nu

- Time-domain clusters do not exceed the null in the 10
  must be cautious.

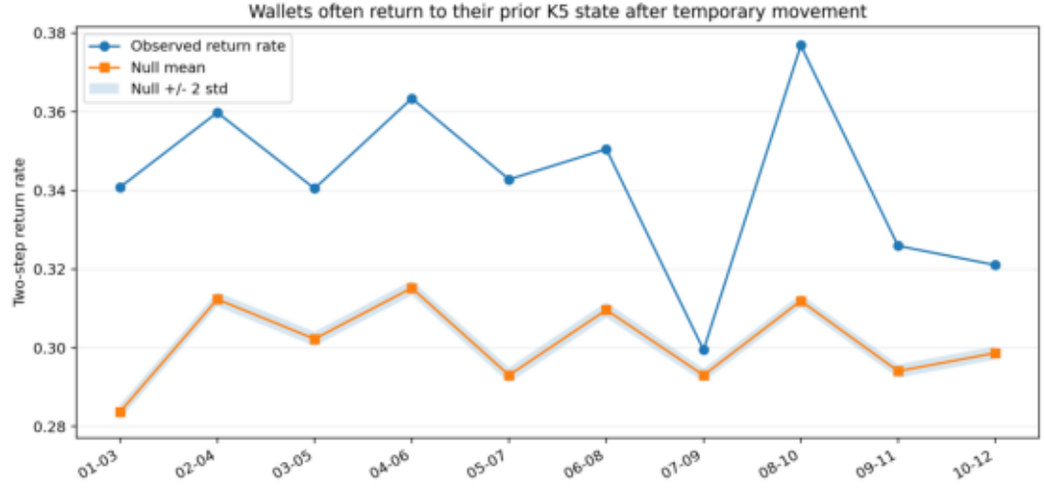


Temporal transitions show temporary movement and return

- Monthly K5 transitions show moderate persistence rather than fixed identities.

- Two-step return rates are consistently above null expectations

- Wallets may temporarily leave a state and later return intent.



The paper contribution is a validation framework

- Layer 1: static aggregate roles from EOA wallet features.
- Layer 2: temporal rhythms from frequency-domain signals.
- Layer 3: burst/gap/persistence states from time-domain signals.
- Layer 4: transaction-network evidence through shuffled-label null tests.
- Layer 5: temporal stability through transition and two-step return tests.

Next work should strengthen claims, not multiply outputs

- Finalize paper figures from the audited figure pack.

- Manually review role labels before writing final public descriptions.

- Use formal event-study methods if causal event language is desired.

- Scale temporal signal assignment beyond the sample if compute budget allows.